

Project Proposal: IT Inventory Tracker

Project Title: IT Inventory Tracker

Course: CS487 – Object-Oriented Design and Analysis

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Introduction

This project proposes an IT Inventory Tracker: a system that helps a university IT department manage its hardware, software licenses, and classroom equipment. It records what the department owns, where each item is, the status of the PCs and monitors in every classroom, orders placed, class schedules, the projects and tasks assigned to staff, and every issue reported, from purchase to retirement.

The system has one kind of user: IT department staff, including supervisors and student assistants, all with the same access. Every user can track the department's items and classroom equipment, create and assign projects and tasks, and log and follow issues.

Problem Statement

A university IT department keeps track of many things at once: equipment and supplies, orders for new items, reported issues, class schedules and the classroom equipment they rely on, and the projects and tasks of its staff. These records are usually spread across spreadsheets, emails, and paper notes. They fall out of date, items get lost, orders are forgotten, issues are reported twice or never resolved, and recurring projects are planned from scratch each time.

This wastes time and money and slows the response when a classroom needs working equipment before a class. A single system gives the department one accurate, searchable record of its items, orders, issues, schedules, and work.

Objectives

- Manage hardware assets from purchase to retirement.
- Track where every item is and what condition it is in, including classroom PCs and monitors, and record items lent out.

- Follow each order for new items from request to delivery.
- Maintain class schedules so staff know which rooms need working equipment and when.
- Organize work into projects and tasks, start recurring projects from templates, and assign tasks to team members.
- Notify the team automatically about scheduled work, for example sending the morning shift its classroom-check project and to-do list.
- Record, update, and resolve equipment issues.
- Keep a complete audit history of every change.
- Require users to log in so each change can be traced to a person.
- Demonstrate object-oriented principles and design patterns throughout the design and code.
- Maintain a clear communication inside the IT team

Proposed Features

The system will include the following features:

1. Asset management (add, update, retire, checkout, checkin)
2. Order tracking (requested, ordered, received)
3. Classroom status tracking (condition of each PC and monitor, per room)
4. Class schedule management (which rooms are in use, and when)
5. Location management (buildings, rooms, storage)
6. Project and task assignment
7. Project templates for recurring work
8. Issue logging and tracking (report, update, resolve)
9. Search and filtering
10. Audit history of every change
11. Reports on asset status, classroom status, orders, open issues, and task progress
12. User login
13. Notifications
14. Scheduled task and Calendar

Object-Oriented Design

The main classes may include:

Class	Responsibility
Asset (abstract)	Common data for all assets: ID, tag, status, location, purchase date
HardwareAsset	Physical items such as PCs, monitors, and network devices
SoftwareLicense	License key, total seats, assigned seats, expiration date
User	Department staff member (supervisor or student assistant): ID, name, login
Order	A request for new items: item, quantity, vendor, status (requested, ordered, received)
Classroom	A room and the PCs and monitors in it, with their status
ClassSchedule	The course, room, days, and times a class meets
Location	Building, room, or storage area
Project, Task	Work assigned to users, with status and due date
ProjectTemplate	A reusable list of tasks used to start a new project
Issue	A reported problem with an asset or classroom: reporter, status, resolution
AuditLog	Records every change: who, what, when, old and new values
InventoryService	Coordinates operations such as status updates, orders, task assignment, and issue logging
Report (abstract)	Base for asset, classroom, order, issue, and task reports

The project will demonstrate the four object-oriented principles:

- **Encapsulation**

- **Inheritance**
- **Abstraction**
- **Polymorphism**

Design patterns under consideration include Factory (creating assets by type and projects from templates), Observer (notifying users of new tasks and issues), Strategy (report generation), and Repository (data access).

Relationships between classes will be shown in UML use-case, class, state, and sequence diagrams, with an ER diagram for the database.

Scope of the Project

The project will focus on the core asset-tracking operations of a single IT department, including login authentication and a notification service for users. It will not include:

- Hardware integration, such as barcode or RFID scanners
- Automatic network discovery of devices
- Integration with vendor portals
- A mobile app

Proposed Technology

- **Programming language:** Python 3.12
- **Storage:** SQLite database
- **Testing:** pytest
- **Design tools:** UML diagrams (PlantUML or draw.io)
- **Version control and project management:** Git and GitHub (Issues, Projects, pull requests)

Expected Outcome

At the end of the semester, the project will deliver a working prototype of the IT Inventory Tracker that supports the core use cases: managing assets, tracking orders, classroom equipment, and class schedules, assigning tasks from project templates, logging issues, searching, and viewing audit history and reports, notification.

The project will also deliver complete design documentation (requirements, user stories, UML, ER, state, and sequence diagrams) that shows how object-oriented analysis and design solve a real-world problem.

Work Plan

The schedule follows the course deliverables. Implementation starts after the system design and runs in parallel with the later diagrams.

Due	Deliverable	Main work
Sep 30	SP1 – Project Proposal	Proposal, team roles, GitHub repository setup
Oct 7	SP2 – Requirements Collection	Functional and non-functional requirements, use cases, use-case diagram
Oct 14	SP3 – User Stories	User stories with acceptance criteria
Oct 21	SP4 – System Design	Architecture, package structure, technology decisions
Oct 28	SP5 – Object Design (UML)	Class diagram, design patterns; start implementation
Nov 4	SP6 – DB Design (ER)	ER diagram, database schema; data layer
Nov 11	SP7 – State Diagram	Asset lifecycle state diagram; classroom status, task, and issue logic
Nov 18	SP8 – Sequence Diagram	Sequence diagrams for key use cases; remaining features
Dec 2	Project Presentation	Slides and live demo
Dec 7	Final Report	Combined design documentation, testing, lessons learned
Dec 9	Implementation Code	Final tested code and run instructions

Conclusion

The IT Inventory Tracker is a practical, well-scoped project for applying the concepts of Object-Oriented Design and Analysis. Its domain has natural class hierarchies, clear object states, and real business rules, which make it a good fit for UML modeling and design patterns. The project will give the team experience in requirements analysis, UML modeling, class and database design, implementation, testing, and team collaboration through GitHub.